

LatentLearn

A Spatial, Non-Linear Learning Space for Divergent Minds

Tech Stack: Next.js / FastAPI / Stateful LangGraph / Google AI Studio / Gemma 4

Target: 2026 AI Agent Innovation Pitch & Demo

01. The "Linear Trap" of Traditional AI Interaction

Waterfall chat threads (like standard ChatGPT) create severe cognitive bottlenecks for associative learners:

1. Context Pollution

When exploring tangents or details, the conversational history becomes cluttered with secondary, off-topic details.

Consequence: The LLM's understanding of the primary subject degrades due to noise in the prompt window.

2. Losing the Trunk

Returning to the core subject after a deep dive requires massive scrolling and a high cognitive load.

Consequence: Frequent disruption of flow state, causing users to get lost and abandon study sessions.

02. The Spatial Paradigm: Breaking the Linear Trap

We represent the learning journey as a hierarchical "Focus Tree" on a spatial, interactive canvas

Core Philosophy

Divergent minds (ADHD or associative thinkers) learn in branches, not straight lines.

We empower users to **spawn infinite sub-branches** for tangential curiosity and **collapse them cleanly** to return to the core trunk.

Inline Branching

Highlight any word in AI responses to instantly spawn a localized sub-node card without cluttering history.

Cognitive Guardrails

Background drift detection alerts you when tangents go too far, offering a 1-click return prompt.

03. Interaction Experience & UI Aesthetics

✨ 1. Premium Glassmorphic UI

Frosted glass card aesthetics (`backdrop-filter`) with responsive light/dark theme support.

Micro-animations and smooth zoom transitions provide highly tactile user feedback.

📌 2. Touchpoint Context Menu

Dragging to highlight words instantly triggers a floating popover menu with `Ask`, `Explain`, and `Expand` options, anchoring curiosity contextually.

🛡️ 3. Drift Inheritance & Reset

Sub-branches of an already off-topic parent node inherit the off-topic status. Canvas alerts present a **"Refocus"** trigger to spring back to the nearest core trunk.

✅ 4. Resolution & Decluttering

Clicking "Got it " collapses and fades out tangential branches. The canvas auto-focuses back to the parent node, maintaining clean, clutter-free spatial environments.

04. Technical Architecture: Decoupled & Asynchronous

Next.js Frontend

Built with **React 18** and **TypeScript 5**.

- Custom **Server-Sent Events (SSE)** streaming parser.
- Local state reducers handle tree node spawning, folding, and camera pan auto-focus.

FastAPI Web Gateway

Asynchronous endpoints maximize concurrency.

- **Security Sandbox:** Cleans input via strict sanitization to block prompt injections.
- **Real-time Streams:** Establishes persistent SSE pipelines.

LangGraph Agent Engine

Stateful multi-agent DAG orchestration.

- **Checkpointing System:** Saves and resumes chat states on specific nodes.
- **Google AI Studio:** Securely schedules Google's premium Gemma 4 model.

05. The Multi-Agent Brain: Stateful LangGraph DAG

Stateful nodes coordinate intent routing, structural query decomposition, context anchoring, and off-topic evaluations

 LangGraph DAG

1. ``intent_router`` (Intent Classifier)

Classifies user query to trigger the correct pipeline.

2. ``decomposer`` (Query Decomposer)

Splits compound queries into sequential sub-questions.

3. ``tutor`` & ``anchor_locator``

Anchors source context and streams markdown responses.

06. Core Technical Breakthroughs



1. SSE Stream Token Filtering

The Pain: When intermediate agents (decomposer, tree-writer) output structured JSONs, standard streaming dumps JSON code directly into user chat bubbles.

The Solve: Engineered an **Event-Stream Node Whitelist filter**. Uses underlying ``astream_events`` to intercept tokens, passing to frontend ONLY when the event is of type ``on_chat_model_stream`` and generated by the `tutor` node.



2. Drift Inheritance & Early Exit

Off-Topic Inheritance: Sub-questions asked on a parent node that is already off-topic automatically inherit the off-topic flag, eliminating expensive LLM validation runs.

Early Exit Optimization: Under decomposition mode, once ``anchor_locator`` flags sub-questions, the state machine triggers an **Early-Exit short-circuit**, bypassing the expensive tutor generation to immediately dispatch the structured QuestionPlan.

07. Developer Experience: Visual Traces & Telemetry



1. LangGraph Studio Local GUI

Fully integrated with **LangGraph CLI tooling**.
Running `langgraph dev` launches an interactive visual workbench:

- **Live Execution Path:** Watch node highlights shift in real-time as state moves through the graph.
- **State Time-Travel:** Interactively modify, rewind, and re-run state variables to mock edge-cases and test errors.



2. LangSmith Observability

Production-grade observability through **LangSmith Deep Tracing Integration**:

- **Call-level Tracing:** Inspect inputs, outputs, system prompts, latency, and exact token costs for each step.
- **Data Feedback Loops:** Group runs as good/bad to construct regression test suites, systematically optimizing prompts.

08. The Intelligent Learning Pipeline

Core Pillar	Backend Agent Logic	Intelligent Mechanism	Cognitive UX Value
1. Query Decomposition	• <code>decomposer</code> node • Structured QuestionPlan schema	• Early-Exit optimization • User intent classifier	Avoids information dump; Splits compound questions sequentially
2. Dynamic Branching	• <code>tree_writer</code> node • Anchor locator coordinate maps	• Local state reducer synchronization • Precise quote anchoring	Prevents vertical scroll chaos; Preserves exploration history
3. Off-Topic Guardrails	• <code>offtopic_eval</code> node • Stored document-summary check	• Drift status inheritance • 1-click refocusing guardrails	Detects topic drift in real-time; Gently guides focus back to core

Let thoughts grow freely, discover depth in connections.

"The mind is not a vessel to be filled, but a fire to be kindled." — Plutarch

Thank you for your watch!